Walid Chtioui

Simulation Software Engineer

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resume in German: walcht/walid_chtioui_lebenslauf.pdf

Professional Experience

Automotive Simulation Engineer

04/2023 – 04/2025 **AR-HUD Simulation Software Engineer**, *CARIAD - Volkswagen AG*, C++ | Python | TS/JS | Lua | CMake | OpenGLES | SomeIP | NDS | DeckGL | Protobuf | GIS

Worked with the AR-HUD team to build tools that helped in identifying and addressing AR-HUD-related issues for Audi/Porsche car models. Showcase video here.

- Built a highly-portable, web-based 3D visualization tool to replay and analyze dumped car data which allowed the AR-HUD team to analyze data from a multitude of sensors (GNSS, ADAS, LD, NDS-based navigation data, etc.) against global satellite imagery.
- Used said visualization/simulation tool to write PoCs which helped in identifying edge cases and writing better requirements.
- Helped AR-HUD providers by providing documented C++ usage-examples on how to efficiently use certain services/interfaces.
- Contributed to identifying and solving issues related with certain AR-HUD functionalities. E.g., Audi's global drone positioning improvements.
- Contributed to writing requirements about navigation-related functionalities.
- Contributed in identifying issues with AR-HUD providers' C++ source code base.
- Worked within a highly interdisciplinary team involving software engineers, product managers, and test engineers.

Freelance/Open-Source-Software Contributor

08/2025 – Present **GPU/Graphics Software Engineer**, C++ | CMake | Git

- Contributing BCn with rate distortion optimization (RDO) GPU compression to Khronos texture format (KTX). PR: KTX-Software #1167
- Contributing to add KTX format reading/writing support to OpenImageIO (OIIO) (which in turn adds KTX support to Blender3D). PR: OpenImageIO #5185

Master Thesis

Title *Direct Volume Rendering of Large Volumetric Datasets in Immersive Environments*

Supervisors Prof. Dr. Christoph Heinzl, Alexander Gall repo: walcht/com.walcht.ctvisualizer

Projects

06/2025 – Present **IP Sockets Network Programming Guide**, walcht/guides/unixs-sockets.html, C | IP Sockets | TCP | UDP | Git

A brief guide into modern network programming using IP sockets on Linux.

12/2024 – Present **High Performance CT/MRI Visualizer**, walcht/com.walcht.ctvisualizer,
C# | C++ | Python | GLSL | CMake | Vulkan | Unity3D | DICOM | Git | Github Actions

A Unity3D plugin for efficiently visualizing and manipulating very large (in the range of 100GBs) CT/MRI volumetric datasets. Project includes subprojects: walcht/cvds, walcht/UnityVulkanPlugin, and walcht/com.walcht.unityd3.

02/2023 – Present **NeoVim Unity IDE**, walcht/neovim-unity,
C# | Lua | Mono | IPC | IP Sockets | DAP | LSP | Git | Github Actions

A set of projects to provide Visual-Studio-like experience for Unity development using NeoVim on Linux and Windows (with WSL2 support) under permissive licenses (MIT). Project includes subprojects: walcht/com.walcht.ide.neovim, walcht/LSP-TCP-socket-adapter, and walcht/unity-dap.

10/2025 – Present **Language Server Protocol TCP Socket Adapter**, walcht/LSP-TCP-socket-adapter,
C# | IP Sockets | WSL2 | Windows | Git | Github Actions

TCP socket adapter for Language Servers (LS) that do not provide a TCP socket endpoint.

Education

10/2022 – 07/2025 **M.Sc. Computer Science**, ***University of Passau***, Passau
Pursued a double Master's degree in Computer Science.

09/2020 – 07/2025 **Engineering Diploma**, ***National School of Computer Science***, Tunis, *Top 4%*
Relevant coursework: Software Development, Operating Systems, Software Design Methodology

09/2018 – 06/2020 **Preparatory Institute for Engineering Studies**, *Tunis*, *Top 7%*
Relevant coursework: Analysis, Linear Algebra, Physics, Control Theory, Probability

Computer Skills

Prog. Languages	C++, Python, C#, TS/JS, C, Lua, GLSL, HLSL
Technologies	Unity3D, .NET (dotnet), Unreal Engine, Git, Protobuf, Docker, DeckGL, D3.js, Three.js
Operating Systems	Linux (Ubuntu), Android Automotive, AOSP
Standards	DICOM, AUTOSAR, NDS, C++20
APIs	Vulkan, OpenGL Core, OpenGL ES
GUIs	Qt, QML, Dear ImGui, tkinter
Management Tools	Jira, DOORS, Confluence
Miscellaneous	Blender, GIS, IP Sockets, Qemu, VirtualBox

Languages

English	Fluent, C2
French	Fluent, C1
German	Beginner-intermediate, B1
Arabic	Fluent, Native

Passau, August 16, 2026